



BARNABUS CARE

Clinical Evidence

Candidate Assessment

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All data in this package is synthetic. It must never be used for diagnosis, treatment, or patient care, and no conclusion drawn from it describes any real patient, clinician, or hospital.



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Read this before you open the data

This assessment cannot be completed in the time allowed. That is deliberate. What you choose to prove, what you refuse to claim, and how visibly you say so are the first things the panel will score.

There are three kinds of requirement. Hard gates are binary and are checked by re-running your code and your analysis against a sealed dataset you will not see; failing any one ends the evaluation regardless of everything else. Scored requirements are weighed against the rubric published in section 10. Ceiling items, marked in the text, are not expected of anyone in twenty hours; they exist so that exceptional work has somewhere to go. One completed ceiling item counts for more than any quantity of ordinary work.

What the panel is actually measuring

Whether you can tell the difference between a number that is true and a number that is merely produced, and whether you will say so when leadership wants the second one.

Whether the causal, experimental, and evaluation methods you have used commercially survive contact with a clinical setting where randomization is constrained, outcomes are censored, the intervention changes clinician behaviour, and a wrong claim can reach a patient.

Whether you are an engineer as well as a scientist. Half of this assessment is code that must run, at a data volume where a correct method implemented carelessly will not finish, and where the pipeline defects are the reason the existing numbers are wrong.

1. Purpose of this assessment

Barnabus Care is evaluating whether you can own clinical evidence, experimentation, and model evaluation. The person in this role decides what Barnabus is permitted to claim about its systems, designs the studies that support those claims, and holds the line when a commercial or clinical stakeholder wants a stronger statement than the evidence allows.

The role is half scientist and half engineer. The person in it writes the pipeline that produces the evidence, owns its correctness under late data, replays, and schema change, and ships the services that evaluate and monitor models in production. A brilliant analysis that exists only in a notebook is not a deliverable in this role.

This is not a modeling exercise. Model accuracy is a minor part of the score. The role owns whether a claim is defensible in front of a regulator, a hospital research committee, and a clinician who will act on it. Final medical interpretation and clinical sign-off remain with authorized clinicians and Barnabus medical governance; part of what is assessed is whether you understand and will defend that boundary.

2. Assessment format

Calendar window	Seven calendar days from receipt.
Maximum working time	Twenty hours. Submit a time log. Your commit history and file timestamps will be compared against it. More time earns no credit; an inaccurate time log is an automatic fail.
Submission	One repository, one evidence and claims memo, one analysis plan, one results report, one evaluation report, one adversarial analysis, one leadership plan, one presentation, one AI-use disclosure, one reproducibility manifest, one time log, three open-problem answers, one critique.
Sealed re-run	The panel will run your submitted pipeline unchanged against a sealed extension of the dataset in which two of the supplied effects are absent by construction. Estimates that move when they should not, or fail to move when they should, are treated as evidence that the pipeline is fitting the sample rather than measuring the effect.
Live defence	Four hours with a cross-functional panel, two ten-minute breaks, including a live engineering review of your running code. Three segments are unprepared and are scored on reasoning. One requirement changes during the session.
Data	Synthetic only. No real patient information is included or permitted.
Technology	Python 3.12 with a pinned environment. Docker required. A SQL engine is supplied; you may substitute one that runs in the container. Notebooks are permitted for exploration only: no reported number may originate in a notebook, and no notebook may appear in the reproduction path.
Scale	The raw event log contains approximately four hundred million rows across twelve months and does not fit in memory on the evaluation machine, which has sixteen gigabytes of RAM and eight cores. Any step that requires loading it whole will not finish. The full pipeline must complete in under forty minutes on that machine.
Services	Two runnable services are required, described in sections 6.7 and 6.8. Both must build and start from a clean checkout with one command each.

3. Scenario

The Barnabus Care preoperative readiness system has been running in silent mode at two hospital sites for nine months. Silent mode means it produced assessments that were recorded but not shown to clinicians. Three months ago, site A switched the system on for a subset of surgical services. Site B remains silent.

Leadership now wants to publish and to use in commercial material three claims: that the system reduces preoperative cancellations on the day of surgery, that it reduces the time from referral to readiness, and that clinicians agree with its recommendations in the large majority of cases. A hospital research committee has asked for the evidence. A prospective

customer has asked for the same numbers with a stronger framing. An internal analyst has already produced a deck showing all three claims comfortably met.

You have been asked to determine what can honestly be claimed, what cannot, and what would have to be done to earn the claims that cannot yet be made. The internal analyst's deck, code, and results are supplied to you. It contains errors. Some are listed in section 4; others are not, and at least one supplied artifact is wrong on purpose.

4. Known and suspected defects in the existing analysis

The following have been confirmed or alleged. Each must be addressed. You will not have time to resolve all of them; state which you resolved, which you contained, and which you left open and why.

1. The switched-on services at site A were chosen by the site, not at random. They are the services that were already performing best.
2. Cancellations are compared between periods, but the later period includes a change in scheduling policy that is unrelated to the system.
3. Time from referral to readiness is measured only for cases that reached readiness. Cases that never reached it are excluded.
4. Cases are counted from the date the system first produced an assessment, so patients who were cancelled before an assessment existed cannot appear in the treated group.
5. The analysis adjusts for a variable that is recorded after the system's recommendation and is plausibly affected by it.
6. The overall effect on cancellation is positive, while the effect within every individual surgical service is negative, or the reverse.
7. Clinician agreement is computed only on the cases a clinician chose to review in detail.
8. Agreement is measured as raw percentage with no correction for the rate expected by chance, and the base rate of the recommended action is above eighty percent.
9. The uplift model used to target the pilot is evaluated with a classification metric rather than a metric appropriate to uplift.
10. Fifty-one comparisons were run across sites, services, segments, and time windows. Three are reported. No adjustment for multiplicity is described.
11. The experiment was monitored daily and stopped when the result crossed significance. No sequential procedure is described.
12. Clinicians at site A discuss cases with each other and cover for one another across services, so treated and untreated cases are not independent.
13. The effect is largest in the first two weeks after switch-on and decays afterwards.
14. The anomaly-detection monitor fires most often on the smallest segments, and the alerts that follow an extreme week are usually followed by a return to normal.

- 15.** Model performance is reported on a period that overlaps the period used to select the decision threshold.
- 16.** One feature used by the model is populated by a workflow step that only occurs after the outcome is effectively determined.
- 17.** Outcome data at site B is more complete for patients who did well.
- 18.** The LLM-as-judge used to score recommendation quality was prompted with the same model family that generated the recommendations, and the judge's agreement with clinicians was never measured.
- 19.** Cost per case is reported as a mean over a distribution with a long right tail, and the comparison group has a different case mix.
- 20.** The confidence intervals reported assume patients are the unit of analysis, while the intervention was applied at the service level.
- 21.** Event timestamps arrive in three different time zones, one of which is recorded as local time without offset. The analysis buckets them by calendar day.
- 22.** Two of the twelve months contain a daylight-saving transition, and duration calculations across those dates are off by an hour in one direction only.
- 23.** Some events arrive up to nine days late. The pipeline computes daily aggregates once and never revisits them, so recent days are permanently understated.
- 24.** A source table is updated in place when a record is corrected. The pipeline reads it at different times for different steps, so two tables in the same report disagree.
- 25.** The event stream contains duplicates from producer retries. Deduplication is by row hash, but the payload includes an ingestion timestamp, so duplicates are not identical and are not removed.
- 26.** Events for a single case can arrive out of order. Workflow state is derived by taking the last row seen rather than the last row by event time.
- 27.** The schema changed in month seven. A column was renamed and its old values were backfilled with a default rather than left null, which is indistinguishable from a real value.
- 28.** A join between cases and services is one-to-many because a case can be re-booked. The row count after the join is never checked, and outcomes are double counted for re-booked cases.
- 29.** An identifier is a string in one source and an integer in another. The join silently produces no matches for a subset, and those cases are dropped without an error.
- 30.** The pipeline sums a floating-point column across parallel partitions, and the total differs between runs in the eighth significant figure. A downstream comparison uses exact equality.
- 31.** The model is trained on features computed from the full historical table and served on features computed from a live view with different null handling.
- 32.** The natural-language-to-SQL assistant executes generated queries against the analytics database with a role that can read every column of every table, including free-text clinical notes, for every site.

5. Materials supplied

- A synthetic longitudinal dataset covering both sites over twelve months: cases, services, clinicians, timestamps for each workflow step, system assessments and versions, recommendations, clinician actions, cancellations, readiness times, and outcome flags. The generating process is known to the panel and is not disclosed to you.
- Two of the three headline effects are real in the data generating process, at a magnitude the panel knows. One is not present at all. A submission that reports three effects is wrong; a submission that reports two but the wrong two is more wrong.
- A censoring and missingness structure that is not random and is not documented.
- The internal analyst's deck, notebook, and results, including at least one number that cannot be reproduced from the supplied data at all.
- A sample of five hundred recommendation and clinician-action pairs, of which one hundred and twenty carry independent labels from two clinical reviewers who disagree on some of them. No adjudication has been done.
- LLM-as-judge scores for the same five hundred pairs, produced by the internal analyst.
- A data dictionary that is incomplete and, in two places, wrong.
- The three claims as leadership currently wishes to state them, in writing.
- A raw append-only event log of approximately four hundred million rows in partitioned columnar files, delivered exactly as the production system emits it, including the duplicates, late arrivals, out-of-order events, and mid-period schema change described above.
- A mutable operational snapshot of the same domain, taken at an unstated moment, which disagrees with the event log in ways you must characterize.
- The internal analyst's SQL, which runs and produces the headline number.
- A question set of one hundred natural-language analytics questions with reference answers, of which a subset are ambiguous, a subset are unanswerable from the available data, and a subset are attempts to extract information the asker is not authorized to see. The three subsets are not labelled.
- A tenancy and authorization model defining which roles may see which sites, services, and columns.

The dataset, the planted defects, the missingness structure, and the sealed extension are authored and difficulty-rated by the Barnabus clinical intelligence and validation team. Your results report must state which defects you found and how hard you judged each to be; the panel will compare your grading to theirs.

6. Required work

6.1 Evidence and claims memo

A maximum two-page memo to the Founder, CMO, CCIO, and Head of Commercial. It must be readable without the rest of your submission.

- For each of the three claims, state one of: supportable as written; supportable only in a weaker form, which you provide word for word; not supportable with the available data; or not supportable by any analysis of observational data of this kind, and say which.
- State plainly which of the three effects you believe is not present at all, and what convinced you.
- Give your best estimate of each effect with an interval, and state separately how much of that interval reflects sampling and how much reflects assumptions that could be wrong.
- Name the single finding that would reverse each of your conclusions, and how you would go and get it.
- State what Barnabus may say to a prospective customer tomorrow, in language you would sign your name to.
- Identify which questions are yours to answer and which require clinical, privacy, or research-ethics authority.

6.2 Estimand, identification, and assumptions

- For each claim, write the estimand precisely: population, treatment, comparison, outcome, time zero, and the contrast. Ambiguity here invalidates everything downstream and is scored as such.
- Draw a causal diagram for each claim, including the site's selection of services, clinician behaviour, the scheduling policy change, and the mechanism by which the system could plausibly affect the outcome. State the identification strategy that follows from it.
- For every variable you adjust for, state why. For every variable you deliberately do not adjust for, state why, distinguishing confounders from mediators, colliders, and instruments.
- Address time-zero alignment explicitly and show that no case can enter the treated group only by surviving long enough to be assessed.
- Define the target trial you would have run if you could, then state exactly where the observational data departs from it.
- State every assumption required for identification, and for each, whether it is testable, partly testable, or untestable. Untestable assumptions must be named as such in the memo, not buried here.
- **Ceiling.** Provide partial identification bounds for at least one claim under a stated relaxation of the untestable assumptions, and report what the data can establish without those assumptions at all.

6.3 Analysis, estimation, and sensitivity

- Pre-specify your analysis before looking at outcomes, and submit the pre-specification as a timestamped commit that precedes the outcome analysis in your git history. The panel will check the order.

- Estimate each effect using at least two approaches that rely on different assumptions, and reconcile them. Agreement between two methods that share the same untestable assumption is not corroboration; say so if that is what you have.
- Handle clustering correctly. The unit of intervention is not the patient. Report inference that respects this, and state the effective number of independent units, which is small.
- Handle censoring and non-random missingness explicitly, and show what your estimate does under at least two plausible alternative missingness mechanisms.
- Report a quantitative bias analysis for unmeasured confounding: how strong would an unmeasured confounder have to be to move your estimate to null, and is a confounder of that strength plausible in this setting?
- Provide negative control analyses: at least one outcome the system could not plausibly affect and at least one exposure period before switch-on. If your method finds an effect there, report it prominently.
- Correct for multiplicity across every comparison you ran, not only those you report, and disclose the full set.
- **Ceiling.** Reproduce the internal analyst's headline number, identify the exact step at which it becomes wrong, and quantify how much of the reported effect each defect contributes.

6.4 Experiment design for what cannot be claimed yet

- Design the study that would earn the claims that observational data cannot support, under the real constraint that a clinician cannot be blinded and a hospital will not randomize individual patients away from a safety tool.
- Specify the randomization unit, the sequence if the design is staggered, the primary and secondary endpoints, the analysis population, and the pre-registered primary analysis.
- Address interference directly: clinicians share wards and discuss cases, so treatment at one unit affects outcomes at another. State the design's exposure mapping and what your estimate means under interference.
- Provide a power analysis by simulation, not by formula, using the correlation structure observed in the supplied data. Report the minimum detectable effect at the number of units actually available, and state plainly whether the study is feasible at that size.
- Specify interim analysis with a stated alpha-spending or sequential procedure, the stopping rules for benefit, futility, and harm, and who is permitted to see interim results.
- State the ethical and governance path: research ethics board, consent or waiver, data-sharing, and what would require the study to stop.
- **Ceiling.** Provide a design in which inference does not depend on asymptotics, appropriate to the small number of independent units available, and demonstrate its operating characteristics by simulation.

6.5 Evaluation of the model and the LLM components

- Establish clinical ground truth from the supplied labels. Two reviewers disagree and no adjudication exists. Define the adjudication process, report agreement corrected for chance, and state the ceiling this places on any measured accuracy.
- Report agreement between the LLM-as-judge scores and the clinician labels. State whether the judge is fit to be used at all, and if not, say what it may still be used for.
- Address the dependence between the judge and the system being judged, and design an evaluation that does not inherit it.
- Evaluate the recommendation model with metrics tied to clinical consequence, not aggregate accuracy: performance at the operating threshold actually used, calibration with a reliability diagram and a proper scoring rule, decision-curve or net-benefit analysis, and subgroup performance with intervals.
- Where the model is used to target an intervention, evaluate it as an uplift problem, not a prediction problem, and report the appropriate curve and summary statistic.
- Detect and report any leakage in the supplied features, and state which reported performance figures are invalid as a result.
- Define what would have to be true for the model to be re-evaluated after a change, and what change would require the evaluation to start again.
- **Ceiling.** Quantify how much of the measured model performance is attributable to label noise rather than model capability, and state the maximum accuracy any model could achieve against ground truth of this quality.

6.6 Monitoring, drift, and alerting

- Design production monitoring for data drift, population shift, model behaviour, and clinical outcome, and specify what each monitor would have caught among the defects in section 4.
- Replace the current anomaly-detection approach with one that does not fire mainly on small segments, and that distinguishes a genuine shift from regression to the mean and from seasonality.
- State the alert budget: how many alerts per week a human can act on, and what your design does to stay inside it. An alerting system that no one acts on is a failed control, not a partial one.
- Define the thresholds at which monitoring escalates to stopping clinical use, who holds that authority, and how the stop is executed.
- Show how a monitoring signal traces back to a specific model, prompt, policy, and data version.

6.7 Reproducibility and evidence integrity

- One command reproduces every number, table, and figure in your reports from raw data in a clean environment. A number that appears in a report and cannot be traced to code is treated as fabricated.
- Pin the environment, seed everything stochastic, and demonstrate that two runs produce byte-identical results.

- Produce a manifest linking each reported number to the script, the data version, and the commit that produced it.
- Label every simulated, imputed, or assumed quantity as such wherever it appears.
- **Ceiling.** Report the results your pipeline produces on a bootstrap of the analytic decisions themselves, showing how much your headline estimate depends on defensible choices you could have made differently.

6.7 Analytic pipeline

Build the pipeline that produces every number in your reports from the raw event log. It is assessed as production engineering, not as analysis support. It must be correct under replay, and it must finish.

- Model the raw log into an analytic layer with a declared grain for every table, asserted in code rather than stated in prose. Every join must assert its expected cardinality and fail loudly when violated.
- Derive workflow state from event time, not arrival order, and handle out-of-order and duplicate events. Deduplication must survive payloads that differ in metadata but represent the same event.
- Normalize time correctly: mixed time zones, one source without offset, daylight-saving transitions, and durations that cross them. State your convention and prove it with tests over the specific dates in the data.
- Support late-arriving data. A day computed today must be revised when data for it arrives nine days from now, and the revision must be visible rather than silent.
- Make the pipeline idempotent. Re-running any step, or the whole pipeline, over the same window must produce the same output, and incremental execution across twelve months must produce output identical to a single full refresh. Demonstrate this; do not assert it.
- Handle the mid-period schema change without treating backfilled defaults as observed values, and record which rows are affected.
- Enforce data contracts covering type, nullability, range, referential integrity, uniqueness at the declared grain, and freshness. A contract violation must stop the pipeline, not warn.
- Reconcile the event log against the operational snapshot, quantify the disagreement, and state which one your analysis treats as authoritative and why.
- Orchestrate it: dependencies, retries with correct semantics for any non-idempotent step, backfill of an arbitrary window, and partial failure recovery. State exactly what happens if the pipeline fails halfway through month nine.
- Meet the budget: full pipeline under forty minutes on the stated machine, with no step loading the raw log into memory. Report peak memory and wall time per step.
- **Ceiling.** Prove idempotency and replay-safety with property-based tests that generate duplicate, out-of-order, late, and partially-failed event sequences rather than with fixed fixtures. Include the floating-point problem: make reported totals identical across partition counts and explain your approach.

6.8 Services

Two services, each building and starting from a clean checkout with one command, each with tests, a health endpoint, structured logging, and a pinned container.

6.8.1 Evaluation and monitoring service

- Expose the monitoring design of section 6.6 as a running service: it ingests scored cases, computes drift, calibration, and outcome monitors, and emits alerts within the alert budget you defined.
- Expose an endpoint that, for any reported metric, returns the exact data version, model version, prompt version, and commit that produced it.
- Handle a metric arriving for a period already computed, and a backfill that changes a previously emitted alert. State what happens to an alert that should never have fired.
- State and meet a latency budget at the ninety-ninth percentile under a stated request rate, and include the load test that demonstrates it.

6.8.2 Natural-language analytics assistant

A service that answers analytics questions over clinical data by generating and executing SQL. This is the most dangerous component in the assessment and is graded accordingly.

- Generated SQL must execute under a role scoped to the asker's authorization. No generated query may read a column, site, or row the asker is not permitted to see, regardless of what the question asks or what the model generates. This is enforced below the model, not by the prompt.
- The service must refuse rather than guess. An ambiguous question must produce a clarification, an unanswerable question must say so, and an attempt to extract unauthorized information must be refused and logged as such. The supplied question set does not tell you which is which.
- Text in the database, including clinical notes, is untrusted. An instruction embedded in a note must never influence the generated SQL, the executed query, or the answer.
- Enforce resource limits: statement timeout, row limit, read-only transaction, no cross-database access, and a scan-size or cost ceiling. A question that would scan the whole event log must be rejected with an explanation, not executed.
- Evaluate the assistant properly: execution accuracy against reference results rather than similarity to reference SQL, refusal precision and recall on the ambiguous and unanswerable subsets, and a separate figure for authorization violations, where the only acceptable value is zero.
- Report cost and latency per question, and state what you would change to halve the cost without weakening any safety guarantee.
- **Ceiling.** Demonstrate the authorization guarantee by construction rather than by testing: show why no generated SQL can violate it, and support the argument with an adversarial suite of at least fifty attempts, including injection through database content.

6.9 Leadership and operating model

- A 30, 60, and 90-day plan for owning clinical evidence and experimentation. The first ten days must state what you would stop.
- The minimum team for the next six months, the hiring sequence, and what you would not attempt until each hire is in place.
- The standard you would set for what Barnabus is permitted to claim, written as rules another analyst could apply without you.
- How you would handle the internal analyst whose deck is wrong, given that the deck has already been shown to a customer.
- What you would do if the Founder asked you to present the stronger framing to a prospective customer next week.
- One page titled “What I will not sign my name to.”

6.10 Adversarial analysis

Attack your own conclusions. Maximum three pages.

- Construct the strongest good-faith case against each of your own estimates.
- State the analysis someone could run on this same data to produce the opposite conclusion, and why it would be wrong.
- Identify the assumption in your work that you are least comfortable with, and what you would do about it with two more weeks.
- Attack the two services as well as the analysis: at least five attempts against the assistant’s authorization boundary and at least three against pipeline correctness under replay, duplication, or late data.
- A submission that finds no serious weakness in its own analysis or its own code is treated as incomplete.

6.11 Open problems

One page each. None has a settled answer. Scored on reasoning and honesty about what cannot be known.

- P1.** The system cannot be randomized away from a clinician who believes it protects patients, and it will never be switched off at a site that has adopted it. Design the strongest possible evidence for a causal claim in a world where the counterfactual is permanently unavailable, and state the point at which you would tell leadership that no study can settle the question.
- P2.** Clinician agreement is used as evidence that the system is good. But clinicians who use the system are influenced by it, so agreement rises as the system's influence grows, whether or not it is right. Describe how you would measure whether the system is correct rather than merely persuasive, and what you would do if the two cannot be separated with the data available.
- P3.** Two hospitals will each accept a different definition of a successful outcome, and the definitions produce opposite conclusions about the same system. You must publish one number. Explain what you publish, what you refuse to publish, and how you defend that choice to whichever site is disadvantaged by it.

- P4.** Every number Barnabus has published to date was produced by a pipeline whose defects you have now documented. Some of those numbers are in customer hands and in a hospital committee submission. Decide which historical results must be recomputed, which must be retracted, and which may stand, and give the rule you would apply rather than the answer for each case.

6.12 Critique this assessment

One page: what this assessment measures that it should not, what it fails to measure, any errors or inconsistencies in the supplied materials, and what you would change. The panel already knows at least two of the answers, and one of them is in this document.

7. Required submission structure

Item	Required content	Limit
01 Evidence and claims memo	Verdict on each claim, the absent effect, estimates with intervals, reversing findings, approved customer language.	2 pages
02 Analysis plan	Estimands, causal diagrams, identification, assumptions and their testability, pre-specification.	8 pages
03 Results report	Estimates by at least two approaches, clustering, missingness, bias analysis, negative controls, multiplicity, defect grading.	10 pages
04 Study design	Design, randomization unit, interference, simulated power, sequential monitoring, ethics and governance path.	6 pages
05 Evaluation report	Ground truth and adjudication, judge validity, threshold and calibration, uplift, subgroups, leakage.	8 pages
06 Monitoring design	Drift, anomaly, alert budget, escalation and stop authority, version traceability.	4 pages
07 Repository	Pipeline, both services, containers, tests, pinned environment, seeds, one-command reproduction, manifest.	No page limit
07a Engineering note	Pipeline design, grain and contracts, idempotency and replay argument, performance budget with measured wall time and peak memory per step.	6 pages
07b Assistant report	Authorization model and enforcement, refusal behaviour, evaluation results including execution accuracy and authorization violations, cost and latency.	5 pages
08 Adversarial analysis	Case against your own estimates, opposite-conclusion analysis, weakest assumption.	3 pages
09 Leadership plan	30, 60, 90 days, team and sequence, claim standard, the analyst, the Founder request, what you will not sign.	6 pages
10 Presentation	Verdict, evidence, what cannot be claimed, plan, and the one thing you need from leadership.	12 slides
11 AI-use disclosure	Tools, where used, material prompts, verification performed, limitations.	2 pages

Item	Required content	Limit
12 Reproducibility manifest	Every number traced to script, data version, and commit.	2 pages
13 Time log	Time by workstream and what you deliberately did not complete.	1 page
14 Open problems	Three answers.	3 pages
15 Assessment critique	What is wrong with this assessment.	1 page

8. Rules and automatic disqualifiers

- Synthetic data only. Do not include any information from a real patient, employer, or client.
- Do not report a number you cannot reproduce, and do not present a simulated or assumed quantity as measured.
- External AI tools are permitted for the take-home and must be disclosed. The live defence is unassisted. Any artifact you cannot explain at the keyboard is treated as fabricated.
- Do not exceed page limits. Content beyond a limit is not read.
- Do not run the outcome analysis before committing the pre-specification. The git history is checked.
- Do not report a confidence interval whose stated coverage your own design does not support.
- Do not widen the assistant's database role, disable a data contract, or reduce the dataset to make a step finish. Every one of these is checked.
- Do not commit the raw data, generated credentials, or any model provider key.

Automatic disqualifiers

Any hard gate in section 10 failing. Any unreproducible reported number. Any claim stated more strongly in the presentation than in the memo. Any inaccurate time log. Any real personal information. Any authorization violation by the analytics assistant. Any prepared material used in a segment marked unprepared. Reporting three positive effects.

9. Live defence

Four hours with two ten-minute breaks. Fifteen minutes of presentation; the rest is challenge. The sealed re-run results are shown to you at the start.

Duration	Segment
15 minutes	Presentation. What can be claimed, what cannot, and what would earn the rest.
30 minutes	Identification challenge with the CMO and CCIO. Your causal diagrams, your untestable assumptions, and your defect grading, contested line by line.
30 minutes	Live analysis review with you at the keyboard. Expect to be asked to re-run an estimate under a changed assumption and to explain a result from the sealed re-run that you have not seen before.
35 minutes	Live engineering review. Your pipeline and services are read and run in front of you. You will be given a corrupted batch to replay, asked to explain a specific query plan and why it is not slower, and asked to defend the authorization boundary against attacks the panel supplies.
20 minutes	Find the flaw. You are handed a second analysis, produced independently, that reaches a different conclusion from yours on the same data. Find what is wrong with it, or concede that it is right.
40 minutes	Live problem. A question with no settled answer, related to but not identical to the open problems, worked at a whiteboard with the panel. Confident language without an argument scores zero.
20 minutes	Commercial pressure simulation. A prospective customer meeting in which the strongest framing is requested in real time and a colleague supports it.
15 minutes	Cold read. A result table you have never seen. State what it supports, what it does not, and what you cannot tell.
20 minutes	Leadership scenario. The wrong deck has reached a customer. You decide what happens next and to whom.
10 minutes	Final statement. What may Barnabus claim, what must it stop claiming, and what authority do you require?

Live change

During the session, one element of the scenario changes: a defect is revealed, a claim is withdrawn, or a constraint on randomization is lifted or imposed. You will state what it does to each of your conclusions and re-run or redesign accordingly.

10. Hard gates and scoring

Hard gates

- G1.** Every reported number reproduces from raw data in a clean environment using only the commands in your README, and two runs are byte-identical.

- G2.** The pre-specification commit precedes the outcome analysis in git history.
- G3.** You correctly identify which of the three claimed effects is not present in the data.
- G4.** Inference respects the level at which the intervention was applied; no patient-level interval is reported for a service-level intervention.
- G5.** No adjustment for a post-treatment variable appears in any primary estimate, and every adjustment set is justified.
- G6.** On the sealed re-run, your pipeline's estimates move as the changed data generating process requires, and do not move where it does not.
- G7.** The memo states at least one thing Barnabus is currently claiming that it must stop claiming.
- G8.** The full pipeline completes on the evaluation machine within the stated budget without loading the raw log into memory, and reports per-step wall time and peak memory.
- G9.** Incremental execution across twelve months produces output identical to a single full refresh, and re-running any window changes nothing.
- G10.** Both services build and start from a clean checkout with one command each and pass their own tests.
- G11.** Zero authorization violations by the analytics assistant across the sealed question set, including questions that embed instructions in database content. One violation ends the evaluation.
- G12.** No reported number originates in a notebook, and no notebook appears in the reproduction path.

Ceiling items

Marked in section 6. Partial identification bounds under relaxed assumptions; reproducing and decomposing the analyst's wrong number; a design whose inference does not rely on asymptotics; the maximum achievable accuracy given label noise; a bootstrap over analytic decisions; property-based proof of replay-safety with deterministic floating-point totals; an authorization guarantee argued by construction. Any one completed and defended raises the top score of its dimension above the published maximum.

Scored dimensions

Dimension	Weight	What earns the top score
Causal reasoning	18	Precise estimands, honest identification, correct treatment of mediators and colliders, time-zero discipline, untestable assumptions named in the memo rather than buried.
Data and pipeline engineering	18	Declared grain and enforced contracts, event-time correctness, idempotent replay proven not asserted, late data revised visibly, within budget at scale.

Dimension	Weight	What earns the top score
Refusal to overclaim	15	Correct verdict on all three claims, customer language that survives scrutiny, willingness to withdraw a claim already in market.
Systems and safety in code	12	Services that build and run, authorization enforced below the model, refusal behaviour, injection resistance, latency and cost under stated budgets.
Statistical execution	12	Correct clustering and multiplicity, credible missingness and bias analysis, negative controls, simulated power, sequential monitoring.
Evaluation and measurement	12	Chance-corrected ground truth, judge validity assessed rather than assumed, threshold and calibration, uplift treated as uplift, leakage found.
Reproducibility and integrity	6	One-command reproduction, deterministic results, complete manifest, simulated quantities labelled.
Leadership under pressure	7	Conduct in the commercial-pressure, cold-read, and live-problem segments; clarity about what requires clinical or ethics authority; handling of the analyst.

The panel will not reward volume, methodological vocabulary, or confident language without an argument. The strongest submission will make it hard for Barnabus to say something untrue, and will say plainly what it did not do.

11. Reference baseline

You may use additional primary sources. At minimum, show that you reviewed the following and state, for each, one concrete thing in your analysis or design that exists because of it.

- [Health Canada, Pre-market guidance for machine learning-enabled medical devices](#)
- [IMDRF, Software as a Medical Device: Clinical Evaluation](#)
- [SPIRIT-AI and CONSORT-AI, reporting guidelines for clinical trials of AI interventions](#)
- [DECIDE-AI, reporting guideline for early-stage clinical evaluation of decision support driven by AI](#)
- [STROBE, reporting of observational studies in epidemiology](#)
- [TRIPOD+AI, reporting of prediction model studies](#)
- [Information and Privacy Commissioner of Ontario, AI guidance for the health sector](#)
- Target trial emulation, quantitative bias analysis, partial identification, and cluster-randomized and stepped-wedge design. Access through your own means; you are expected to apply these without being taught them.

12. Final instruction

The number you refuse to publish

You are not expected to answer every question in the data or to build every component well. You are expected to establish what is true, to build the machinery that keeps it true when the data is replayed, to state clearly what cannot be established at all, and to leave Barnabus with claims it can defend in front of a regulator, a hospital, and a clinician. What you refuse to publish, and what you refuse to ship, are scored as seriously as what you deliver.